Automated Career Assistant

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A Practical Agentic AI System

1. Introduction

The rapid growth of Artificial Intelligence has led to the development of intelligent systems that not only generate text but can also reason, take decisions, and perform real-world actions. These systems, commonly referred to as **Agentic AI**, are capable of orchestrating multiple tools, handling complex workflows, and executing tasks autonomously.

This semester project, titled "Automated Career Assistant", presents a Track A – Practical Agent System that assists job seekers in analyzing resumes against job descriptions, identifying skill gaps, and generating tailored resumes, cover letters, and application emails. The system integrates Agentic AI, Retrieval-Augmented Generation (RAG), and tool-based execution using modern frameworks such as LangChain, FastAPI, and Google Gemini AI.

Unlike traditional chatbots, this system performs **real actions** such as parsing PDF files, generating professional documents, sending emails, and evaluating agent behavior using deterministic benchmarks.

2. Project Objectives

The main objectives of this project are:

- To design and implement a practical agentic AI system
- To apply Retrieval-Augmented Generation (RAG) for grounded AI outputs
- To enable autonomous decision-making and tool execution
- To analyze resumes and job descriptions using AI
- To identify skill gaps and calculate match percentages
- To generate ATS-optimized resumes and professional cover letters
- To evaluate agent performance using deterministic benchmarking metrics

3. Problem Statement

Job seekers often struggle to:

- Understand how well their resume matches a job description
- Identify missing or weak skills
- Customize resumes and cover letters for different roles
- Prepare professional application documents efficiently

Existing solutions are either manual, time-consuming, or lack intelligent automation. This project addresses these challenges by introducing an **agent-based system** that automates the entire job application workflow.

4. Project Scope

The scope of the Automated Career Assistant includes:

- Resume and job description analysis (PDF format)
- AI-powered information extraction
- Skill gap detection and scoring
- Tailored resume and cover letter generation (DOCX & PDF)
- Optional email automation for job applications
- Agent orchestration and evaluation

Out of scope:

- User authentication
- Persistent database storage
- Mobile application support

5. System Architecture

The system follows a layered agentic architecture consisting of the following components:

5.1 Frontend Layer

- Built using **Next.js** and **React**
- Provides user interface for file upload, analysis results, document generation, and email sending
- Communicates with backend through REST APIs

5.2 Backend Layer

- Developed using **FastAPI (Python)**
- Handles file uploads, API requests, and responses
- Manages AI services, document generation, and email sending

5.3 Agent Layer

- Implemented using **LangChain**
- Uses the **ReAct (Reasoning + Acting)** paradigm
- Dynamically selects and executes tools based on user requests

5.4 Retrieval Layer (RAG)

- Retrieves content from uploaded resume and job description PDFs
- Injects retrieved content into AI prompts
- Ensures grounded and factual AI responses

5.5 Tool Layer

- Custom tools for PDF parsing, AI analysis, skill comparison, document generation, validation, and email sending

5.6 Evaluation Layer

- Deterministic agent evaluator
- Benchmarks agent behavior using predefined rules and metrics

6. Agent Description

6.1 Tasks Performed by the Agent

The agent autonomously performs the following tasks:

1. Parse resume and job description PDFs
2. Extract structured information from unstructured text
3. Analyze skills and experience
4. Identify skill gaps
5. Calculate match percentage
6. Generate tailored resumes
7. Generate personalized cover letters
8. Validate generated documents
9. Send job application emails (optional)

These tasks involve **multi-step reasoning, tool selection, and real-world execution**.

6.2 Tools Used

The agent is connected to the following custom LangChain tools:

Tool Name	Description
parse_pdf	Extracts text from PDF files
analyze_job_description	Extracts job details using Gemini AI
analyze_resume	Extracts candidate details using Gemini AI
analyze_skill_gap	Compares resume and job skills
generate_tailored_resume	Generates ATS-optimized resume
generate_cover_letter	Generates professional cover letter
send_application_email	Sends email with attachments
validate_documents	Verifies file existence

6.3 Reasoning Style

The agent follows the **ReAct (Reasoning + Acting)** reasoning pattern:

- Reason about the task
- Select an appropriate tool
- Execute the tool
- Observe the result
- Decide the next step

This approach allows the agent to behave autonomously and adaptively.

7. Implementation Details

7.1 Backend Implementation

- **FastAPI** is used for building RESTful APIs
- All file handling and processing is asynchronous
- PDFs are temporarily stored using unique UUID-based filenames

7.2 AI Integration

- **Google Gemini (gemini-2.5-flash)** is used for:
  - Job description analysis
  - Resume analysis
  - Resume and cover letter generation
- Structured prompts ensure consistent and validated outputs

7.3 Skill Gap Analysis Algorithm

The skill matching algorithm categorizes skills into:

- Matching Skills (exact matches)
- Partial Skills (related skills)
- Missing Skills (required but absent)

Match Percentage Formula:

(Matching + 0.5 × Partial) / Total Required × 100

7.4 Document Generation

- **python-docx** is used for DOCX generation
- **ReportLab** is used for PDF generation
- Markdown parsing preserves formatting
- Professional styling is applied

7.5 Email Automation

- Uses **SMTP (Gmail)**
- Sends multipart emails with attachments
- Credentials are securely managed via environment variables

8. Agent Evaluation

8.1 Evaluation Approach

A **Deterministic Agent Evaluator** is implemented to ensure:

- Reproducibility
- No LLM dependency during evaluation

- Objective scoring

8.2 Benchmarking Metrics Used

The agent is evaluated using **seven metrics**:

1. Task success rate
2. Action execution correctness
3. Tool usage correctness
4. Agent task accuracy (Graphene-style)
5. Reasoning length
6. Latency
7. Cost efficiency

8.3 Test Scenarios

Eight predefined test scenarios are executed, including:

- Resume parsing
- Job analysis
- Skill gap computation
- Resume generation
- Cover letter generation
- Full workflow execution
- Error handling
- Validation-only tasks

Each scenario produces PASS / PARTIAL / FAIL results with evidence.

9. Security and Best Practices

- API keys stored in `.env` file
- `.gitignore` prevents credential leakage
- UUID-based file naming
- Automatic file cleanup
- Robust error handling and logging
- CORS configuration for restricted access

10. Demo and Usage

The demo includes:

- Uploading PDFs
- Viewing analysis results
- Generating documents
- Sending emails
- Running agent evaluation

11. Future Enhancements

- User authentication
- Database integration
- Job portal integration
- Interview preparation module
- Multi-language support
- Mobile application

12. Conclusion

The Automated Career Assistant successfully demonstrates a **Track A Practical Agent System** by integrating Agentic AI, RAG, tool orchestration, and deterministic evaluation. The system performs real-world actions, exhibits autonomous reasoning, and provides measurable outputs. This project highlights proficiency in full-stack development, AI integration, and advanced agent-based system design.